

COS341 Project Phase 1: SPL Front End

User Manual of Group 31

1 Group members

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2 Requirements

The executable is the Java archive **group-31.jar**. It needs Java 8 or newer and nothing else. Typing `java -version` in a command prompt shows the installed version. The same jar works on Windows as well as on Linux and macOS.

3 Running the program

1. Extract the zip file into a folder.
2. Put the test file **SPL.txt** into the same folder as **group-31.jar**.
3. Open a command prompt in that folder and run the jar.

```
java -jar group-31.jar
```

Please start the jar from a command prompt. With a double-click Windows shows no console window and the messages stay hidden.

Another input file can also be given as an argument. Paths with spaces need double quotes.

```
java -jar group-31.jar C:\tests\case3.txt
```

Without an argument the program looks for **SPL.txt** in the current folder first. If it is not there the folder of the jar is tried next.

4 Output for a correct program

For a syntactically correct program the file **tree.xml** is written into the folder of the input file. The program prints its full path. A web browser can display it.

```
SPL.txt contains a syntactically correct SPL program.  
The syntax tree has 61 nodes and was written to C:\COS341\tree.xml
```

The file follows the layout of the specification. Its three parts are the root, the inner nodes and the leaves.

Element	Meaning
UNID	unique ID number of the node
SYMB	non-terminal of the root or an inner node
TERMINAL	token of a leaf as it appears in the input
PARENT	ID of the node above
CHILDREN	IDs of the children from left to right

The root is `SPL_PROG` and its only child is `P`. Since the end marker `$` is not part of the input file the tree has no leaf for it. A non-terminal that derives the empty word shows up as an inner node without children.

5 Output for an incorrect program

No `tree.xml` is produced in this case and an older one in that folder is deleted. Instead the program prints the line and column of the first error together with a hint. A lexical error means that some text is not a valid token. With a syntax error every token is valid but their order does not fit the grammar.

```
Syntax error in SPL.txt on line 7, column 1:
Found '}' but expected ';' to end the instruction.
```

```
7 | } ;
   | ^
```

```
Hint: The last instruction before a '}' needs a ';' as well. The instruction
      that needs it starts on line 6.
```

6 Exit codes

Code	Meaning
0	correct program and <code>tree.xml</code> written
1	lexical or syntax error found
2	input file missing or not readable

7 Troubleshooting

'java' is not recognized as an internal or external command

Java is missing or not on the search path. Install Java 8 or newer and open a new command prompt.

Error: Unable to access jarfile

The command prompt is in another folder. Change to the folder of the jar or give its full path.

Could not find the input file

Check the file name. Windows may hide the extension so a file shown as `SPL.txt` could really be `SPL.txt.txt`.